Quiz #6

1. For a confidence interval for a population mean, determine how the **width** changes in each of the following situations, assuming all other quantities remain constant.

   (a) The sample size n increases.

   (b) The confidence level increases.

   (c) The population mean μ increases.

   (d) The population standard deviation σ increases.

   For each part, choose one of the following answers:

   - Gets wider
   - Gets narrower
   - Stays the same

2. Fill in the blanks using **null** or **alternative**.

   The _____ hypothesis says that an observed difference can be explained by chance variation, while the _____ hypothesis proposes that the difference reflects a real difference in the population.

3. Consider the following P-values:

   $$0.1\%, \qquad 3\%, \qquad 17\%, \qquad 32\%.$$

   (a) Which P-value provides the strongest evidence for retaining the null hypothesis? Explain briefly.

   (b) Which P-value provides the strongest evidence against the null hypothesis? Explain briefly.

4. State whether each statement is **true** or **false** and briefly explain.

   (a) The observed significance level depends on the data.

   (b) If the observed significance level is 5%, there is a 95% probability that the alternative hypothesis is true.

5. A NumPy array named `readings` contains independent measurements from an approximately normal population. The population SD is unknown.

```
from scipy import stats

n = len(readings)

xbar = readings.mean()

se = stats.sem(readings)

t_star = stats.t.ppf(0.975, df=n-1)

ci = (xbar - t_star*se, xbar + t_star*se)
```

   (a) In words, explain what the code calculates. What are `se`, `t_star`, and `ci`?

   (b) Why does the code use `0.975` in `stats.t.ppf` for a 95% confidence interval?